

Beyond Relation Count: Endpoint-Sharing Topology in Text-Guided 3D Scene Generation

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Preconfirmatory Advisor Draft. The primary SGP-14 training/confirmation is still in progress. Development and lightweight qualification are not paper evidence. All result sections below are placeholders tied to a frozen GO/STOP protocol; no topology effect is asserted before the required checkpoints and untouched-test confirmation exist.

Abstract

Multi-relation scene instructions are commonly summarized by relation count, although constraints with the same count may have different endpoint-connectivity structure. We isolate a narrower structural question: under matched objects, relation count, predicate multiset, text budget, and decoder, does concentrating relations around a shared endpoint create a residual difficulty relative to chain-like structure? We construct paired Chain and Hub instructions at counts three and four in an InstructScene-style semantic-graph cascade. The study also uses an exact-identity oracle-graph substitution under identical decoder randomness to localize any residual to at-or-before the text-to-graph interface versus the shared graph-to-scene decoder. Current development work establishes the controlled compiler and training qualification path; the primary topology result remains prospective until new SGP-14 checkpoints and untouched-test confirmation complete. The intended contribution is a count-matched topology diagnostic, not a new claim about generic relation-count degradation or graph-based scene generation.

1 Why relation count may be insufficient

Text-guided 3D scene generators must satisfy several relational constraints simultaneously. Existing evaluations often treat the number of relations as the main complexity axis. Yet three or four relations may form a dependency chain or share a common anchor. These instructions have the same edge count while inducing different endpoint-connectivity structure. If topology contributes difficulty beyond count, then relation-count curves confound two variables and can obscure where failures originate.

The paper therefore asks two controlled questions: (i) after matching relation count and content, does endpoint sharing create a residual generation gap? and (ii) if a residual exists, is it already present at the semantic-graph interface or does it emerge only during graph-to-layout decoding?

2 Closest work and novelty boundary

InstructScene establishes a semantic-graph-prior pipeline for instruction-driven 3D scene synthesis. SceneNAT and related work already study stronger relation-count regimes and beyond-training relation counts. GeoSceneGraph and other graph-guided generators establish that graph structure is useful in scene synthesis. We do not claim novelty for graph priors, relation-count degradation, or graph-stage diagnostics in general.

The defended residual is the intervention itself: *same-count, same-object, relation-composition-matched Chain/Hub counterfactuals* plus *exact-identity predicted-versus-oracle graph substitution under the same decoder*. The contribution exists only if this controlled comparison reveals a reproducible residual and the localization gate behaves as preregistered.

3 Controlled topology intervention

The generation interface is

instruction $\rightarrow$ semantic graph predictor $\rightarrow$ frozen shared graph-to-scene decoder $\rightarrow$ layout.

We compare Chain and Hub structures at relation counts three and four. Count two is excluded from topology evidence because the candidate structures are isomorphic.

Within each matched pair we hold fixed the base scene, active object identities/classes/features/masks, relation count, predicate/direction multiset, instruction template family, exact token-count checks, tokenizer revision, and no-truncation status. The panel compiler is frozen before primary model outcomes are inspected.

4 Exact-identity graph intervention

For each text instruction, two decoder branches receive identical object/slot identity fields and paired decoder randomness:

1. the semantic graph predicted from text;
2. the ground-truth instructed graph.

The oracle branch intentionally receives correct relational information and is therefore an intervention/upper bound, not a fair deployment baseline. If the Chain/Hub residual substantially contracts under oracle substitution while the decoder is otherwise unchanged, the evidence localizes a material component to at-or-before the semantic-graph interface. If the residual persists, the downstream decoder or a later interaction remains implicated.

5 Observables

We report text-to-graph relation recall, conditional graph-to-scene retention, and unconditional end-to-end instruction recall (iRecall). We do not multiply these metrics into a pseudo-causal waterfall. Stage localization comes from the exact predicted-versus-oracle substitution, not from correlating intermediate scores.

6 Training objects and current status

Primary training uses SGP-14 with one frozen shared decoder; the current controlled data geometry contains 128 training and 48 validation pairs per topology under semantics including **behind** and **right_of**. Chain, Hub, and Community are generated, but Community remains exploratory until its prespecified failure rule is cleared. SGP-12 is secondary context only because its relation-edge/token exposure differs and cannot identify support causality by itself.

Object	Current status	Claim authority
Matched compiler	Implemented / qualified for controlled generation.	Supports protocol validity only.
SGP-14 primary checkpoints	Training from scratch under the frozen primary run.	No topology paper claim until required checkpoints complete.
Development/lightweight screens	Available for debugging and GO/STOP preparation.	Not paper evidence.
Community topology	Generated as exploratory structure.	No main claim unless the documented gate passes.

7 Prospective evaluation

7.1 Primary estimand

For each matched physical base scene, define

$$\Delta_{topo} = \text{mean}(\text{iRecall}_{Hub} - \text{iRecall}_{Chain}).$$

We report count-3, count-4, and equally count-weighted pooled effects. Inference uses a physical-base-scene cluster bootstrap with 10,000 replicates; if a scene appears in both counts, all its observations are resampled together.

7.2 Frozen GO/STOP

Validation must satisfy the preregistered direction/quality gates before untouched-test confirmation. A failed validation gate stops topology claims; it does not trigger new topology definitions or outcome-conditioned panel changes.

7.3 Untouched-test confirmation

Conditional on validation GO, two additional independent SGP-14 seeds are trained for three total seeds and evaluated using the unchanged compiler and analysis. Claims remain explicitly conditional on the one frozen shared decoder.

8 Planned results structure

The paper will answer, in order: (1) whether endpoint-sharing topology matters within SGP-14’s training range; (2) whether the same directional difference appears at text-to-graph; (3) whether exact graph substitution reduces the end-to-end topology gap; and (4) whether the effect replicates across independent SGP seeds. Until the frozen primary run is complete, these are questions, not results.

9 Strongest alternatives and hard stops

The design is invalidated or the claim must narrow if relation count, active objects, predicate composition, text length, or decoder randomness are not truly matched; if the effect exists only in a development panel; if the topology gap changes sign across required seeds without a stable pooled effect; or if oracle substitution does not support the proposed localization. A graph-based architecture advantage or relation-count decline alone does not count as evidence for the topology claim.

10 Limitations

The study uses one scene-generation substrate, a small set of relation semantics, and one frozen decoder. Three SGP seeds establish minimal reproducibility rather than a full optimization-randomness distribution. Endpoint sharing is one structural axis, not a universal graph law. Realistic instructions may co-vary relation count, topology, semantics, and object composition; the controlled study deliberately isolates only one factor.

11 Conclusion

This paper tests whether relation topology contributes learnability beyond relation count in text-guided 3D scene generation. The experimental contribution is a matched Chain/Hub intervention plus exact graph substitution for stage localization. Current work establishes the protocol and training path, not the topology result. The paper becomes an empirical claim only after the frozen SGP-14 validation and untouched-test evidence close the preregistered gates.

Related systems used to define the boundary

InstructScene motivates the semantic-graph cascade; SceneNAT motivates relation-count evaluation and extrapolation; GeoScene-Graph and related graph-guided systems establish graph structure as a generation primitive. Exact bibliographic metadata will be normalized before venue submission; the advisor draft uses these systems only to make the novelty boundary explicit.